An Endpoint-Swap Obstruction for Natural Collatz Cycles

Collatz proof project

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Abstract

We formalize an elementary cancellation obstruction for actual natural Collatz returns. If the binary words $1u0$ and $0u1$ are realized by returning natural seeds, then u is empty and the seeds are respectively 1 and 2. No palindrome, coprimality, or common-orbit assumption is required. The argument adapts the cancellation underlying Knight's high-cycle exclusion to the main project's word and orbit definitions. It is not a proof of Collatz or a claim of mathematical novelty. The word-structure bridge needed to apply it to all rational mechanical itineraries remains open in this main-project formalization.

1 Actual return words

Use the shortcut map $T(n) = n/2$ for even n and $(3n+1)/2$ for odd n. A word w is realized by n when its bits give the parities of the first $|w|$ orbit values. Let $j(w)$ count its ones and let $C(w)$ denote its affine correction, defined recursively by

$$C(\epsilon) = 0, \quad C(0w) = 2C(w), \quad C(1w) = 3^{j(w)} + 2C(w).$$

The verified affine identity is $2^{|w|}T^{|w|}(n) = 3^{j(w)}n + C(w)$.

Theorem 1. *Let u be any finite binary word. If x realizes $1u0$, y realizes $0u1$, and both seeds return after $|u| + 2$ shortcut steps, then*

$$u = \epsilon, \qquad x = 1, \qquad y = 2.$$

The first letter of $1u0$ already forces $x > 0$. The two returns need not belong to the same orbit. For one cyclic orbit, the theorem applies whenever a shift realizes the second word; a return at one state implies the same return length at every shifted state.

2 Cancellation proof

Set $\ell = |u|$, $j = j(u) + 1$, $A = 2^{\ell+2}$, and $B = 3^j$. The positive-cycle inequality gives $B < A$, so $D = A - B$ is a positive odd integer. Write $C_L = C(1u0)$ and $C_R = C(0u1)$. Direct expansion, valid for every u, yields

$$3C_L + A = C_R + B + 2^{\ell+1}.$$

The two return identities give $C_L = Dx$ and $C_R = Dy$. Substitution gives

$$D(3x + 1) = Dy + 2^{\ell+1}, \qquad D \mid 2^{\ell+1}.$$

An odd positive divisor of a power of two is 1. Thus $A = B + 1$. If $\ell > 0$, then A is divisible by 8, whereas 3^j is congruent to 1 or 3 modulo 8. This contradicts $A = B + 1$. Consequently $\ell = 0$. The two affine equations now reduce to $4x = 3x + 1$ and $4y = 3y + 2$, giving $x = 1$ and $y = 2$.

3 Lean statements and validation

The new module `Collatz/EndpointCycles.lean` provides

- `endpoint_cancellation`, the arbitrary-word identity;
- `endpoint_returns_trivial`, the two-seed theorem;
- `endpoint_cycle_shift_trivial`, its single-orbit corollary.

All three are included in the selected axiom audit. The module builds with only standard foundational axiom dependencies; the conditional finite-range verification in `Cycles.lean` is not used. The audit is a Lean build and dependency inspection, not an independent kernel replay.

The exact Python controls in `analysis/test_endpoint_cycles.py` check the cancellation and paired-integrality conclusion for every middle of length at most 12. They also find endpoint-swapped rotations for every reduced rational mechanical slope p/q with $0 < p < q \leq 80$. These are finite controls, not a proof of the universal rotation statement.

4 Provenance and remaining bridge

The cancellation was located in the nested repository `tcosmo__Knight2026_lean`. The relevant file is `Knight2026/NoHighCycles.lean`. Its high-cycle proof also uses Christoffel word structure and rotation preservation of integrality. The present proof uses the same mathematical idea, with fresh natural-number arithmetic in the main project's definitions, and imports no nested module. The relevant published source is Kevin Knight, *Collatz high cycles do not exist*, Discrete Mathematics 349(3) (2026), article 114812, doi:10.1016/j.disc.2025.114812. The nested summary's article number 114425 and HAL identifier are inaccurate; the publisher lists 114812 and the challenge catalog links HAL-04261183.

To finish the rational mechanical route, one must prove that a primitive rational mechanical period has both endpoint-swapped words among its rotations, and transport those rotations to actual returning states. Handling nonprimitive periods and eventual tails is also required. The irrational mechanical exclusion already proved in the main project does not supply these steps.

The witness property is not universal for arbitrary parity words. For example, 11100 has no endpoint-swapped pair among its rotations. Its formal cyclic value is 19/5, so it is a rational example illustrating the structural gap, not an integer Collatz counterexample. Neither general nontrivial cycles nor unbounded orbits are excluded by this result.